

MathPinpoint: A Large-Scale Dataset for Problem-Level, Correctness-Aware Math Retrieval

Zefeng Cai

Enrico Shippole

Abstract

A student checking a solution, a tutor preparing feedback, or a language model grounding its answer in retrieved text needs the page that solves *this* math problem, and solves it correctly. Retrieval systems, and most benchmarks used to evaluate them, reward topical similarity instead: a page about the same concept, or about the same problem with one condition changed, counts as relevant. We introduce MathPinpoint, a dataset for problem-level, correctness-aware math retrieval. Every label follows one written rubric that gives full credit only to a page that answers the same problem with correct mathematics. The corpus holds 3.5M deduplicated mathematical web pages. The training split contains 1.85M queries extracted from those pages, either a page’s own question with minimal edits or a question composed from a page that states none, and 43.5M judged candidates from three retrieval routes; each label comes from GPT-5.6-Sol or from an 8B relevance model distilled from it, and records which. The test split holds 6,173 held-out queries whose pooled candidates were judged by two LLM judges and an LLM arbiter, giving 83,833 relevant pairs. Fine-tuning Qwen3-Embedding-0.6B on the training split raises nDCG@10 on the test split from 0.441 to 0.477. Pool bias is large at this scale: the judged pool covers 66% of the base model’s top 10 but 35% of the fine-tuned model’s, and two corrections for it place the gain between +0.11 and +0.13. BM25, whose own results were pooled, has the highest standard nDCG@10 (0.562); on 500 queries whose top-10 rankings were judged in full, the fine-tuned model ties it (0.577 against 0.562). The data is released under CC0, and the prompts, schemas and evaluation code under Apache-2.0.

1 Introduction

Mathematical problems recur across the web. A textbook exercise is posted on a homework fo-

rum, solved on a tutoring site, copied into lecture notes and re-posted with different numbers. Someone holding a specific problem, whether a student checking a solution, a tutor preparing feedback, or a language model that grounds its answer in retrieved text, needs a page that solves *that* problem. Two kinds of near miss make this hard (Figure 1). The first is the page that is about the right topic, or about a nearly identical problem in which one condition, one quantity or the requested target differs; such pages are easy to retrieve and of little use. The second is the page that addresses the right problem but gets the mathematics wrong; it is worse than no answer.

Retrieval benchmarks rarely measure either distinction. General-purpose collections such as BEIR (Thakur et al., 2021) and MTEB (Muenighoff et al., 2023) contain few mathematical tasks, and their relevance labels are topical. Evaluations built for mathematical information retrieval, such as the NTCIR math tasks (Zanibbi et al., 2016) and ARQMath (Zanibbi et al., 2020), assess formula and answer retrieval over community question answering with a small number of human-assessed topics. Reasoning-intensive benchmarks such as BRIGHT (Su et al., 2025) include mathematical subsets, but there a relevant document is one that uses the same theorem or technique, which is a different need from solving the same problem. None of the mathematical ones provides training data at the scale that current embedding models are trained on.

We present **MathPinpoint**, a dataset for problem-level, correctness-aware math retrieval. Its relevance rubric (§3.2) is strict in two ways. A page gets full credit only when it addresses the same problem: a different problem, changed conditions, a different target or shared keywords all score 0. And a page gets full credit only when its central mathematics is correct: a wrong solution to the right problem scores 0, and so does a correct

Query: What is the square of $3 + 5i$?	
2	$(3 + 5i)^2 = 3^2 + 2 \cdot 3 \cdot 5i + (5i)^2 = 9 + 30i + 25i^2 = \dots = -16 + 30i$ <i>dense rank 1 · full_solution</i>
0	$(5 + 3i)^2 = 5^2 + 2(5)(3i) + (3i)^2 = 25 + 30i + 9i^2 = \dots = 16 + 30i$ <i>dense rank 3 · wrong_problem</i>
0	$(3 - 5i)(3 + 5i) = 3^2 - (5i)^2 = 9 - 5^2 \cdot i^2 = \dots = 34$ <i>dense rank 8 · target_mismatch</i>
1	$(2 + 3i)^2 = (2 + 3i)(2 + 3i) = 4 + 6i + 6i + 9i^2 = \dots = -5 + 12i$ <i>dense rank 14 · correct_partial_method</i>

Query: Solve $x(x + 4) = 21$.	
2	$x^2 + 4x = 21; x^2 + 4x + 4 = 21 + 4;$ $(x + 2)^2 = 25; x + 2 = \pm 5; x = 3 \text{ or } -7$ <i>dense rank 4 · full_solution</i>
0	We solve by factoring: $x^2 - 4x = 21;$ $x^2 - 4x - 21 = 0; (x + 3)(x - 7) = 0;$ so $x = -3$ or $x = 7$ <i>dense rank 8 · major_math_error</i>

Figure 1: Judged candidates of two training queries, with the LLM judge’s labels and reason codes (excerpts from the released corpus). Left: the dense retriever ranks a page that squares $5 + 3i$ third and one that multiplies $3 + 5i$ by its conjugate eighth; both answer a different problem and score 0, while a correct method applied to another number scores 1. Right: a page that sets out to solve the right equation but expands $x(x + 4)$ as $x^2 - 4x$ scores 0.

final answer reached through invalid reasoning.

MathPinpoint has three parts. The **corpus** contains 3,525,546 cleaned and deduplicated mathematical web pages from the L2 tier of UltraDataMath (Zhou et al., 2026) (§3.3). The **training split** pairs 1,853,389 queries, extracted from those pages with a minimal-edit prompt that reuses a page’s own question nearly verbatim when it has one (§3.4), with 43,506,169 judged candidates drawn from dense, BM25 and word 5-gram retrieval (§3.5). Each training label comes either from an LLM judge, GPT-5.6-Sol (OpenAI, 2026), or from an 8B relevance model distilled from it, and every row records which (§3.6). The **test split** contains 6,173 held-out queries whose candidate pools were judged independently by two LLM judges, with a third LLM arbitrating disagreements, yielding 83,833 relevant (query, document) pairs (§3.7).

We make four contributions.

- A task definition and a written rubric for problem-level, correctness-aware relevance, with three grades and 18 reason codes, pinned by its SHA-256 (§3.2).
- A dataset at a scale suitable for training retrievers: 43.5M judged pairs, each with the identity of its judge and, where the 8B model scored it, its raw score (§3). We measure the quality of both judges against independent LLM judgments (§4).
- Baselines for lexical and dense retrievers, and a fine-tuning experiment showing that training on MathPinpoint improves a strong embedding model (§5).

- An analysis of pool bias in a benchmark of this size: how much of each system’s ranking the judged pool covers, how two corrections change the measured difference between systems, and what that implies for comparing new retrievers on the test split (§6).

The data is released under CC0 1.0 at <https://huggingface.co/datasets/oklenAI/MathPinpoint>, and the prompts, schemas and evaluation code under Apache-2.0.

2 Related Work

Retrieval benchmarks. MS MARCO (Bajaj et al., 2018) established large-scale passage ranking with sparse human labels. BEIR (Thakur et al., 2021) collected heterogeneous retrieval tasks to test zero-shot generalisation, and MTEB (Muenighoff et al., 2023) and its multilingual extension MMTEB (Enevoldsen et al., 2025) placed retrieval alongside other embedding tasks under one evaluation framework. MIRACL (Zhang et al., 2023) extended human-judged retrieval to 18 languages. Their labels measure whether a document is on topic or answers an information need; none targets mathematical problems. BRIGHT (Su et al., 2025) and RAR-b (Xiao et al., 2024a) evaluate retrieval that requires reasoning. BRIGHT’s mathematical sets count a document as relevant when it shares the query’s problem-solving technique or theorem: another problem solved with the same technique or theorem, or the statement of the theorem itself. MathPinpoint asks a narrower and more literal question: does the page solve this problem, correctly?

Benchmark	Domain	Test queries	Corpus	Relevance	Who labels	Training data
MS MARCO (Bajaj et al., 2018)	web passages	6,980 (dev)	8.8M	binary	search-engine editors, sparse	yes
BEIR (Thakur et al., 2021)	18 datasets, 9 tasks	49–13,145	3.6K–15M	mixed	inherited from the source datasets	some datasets
MIRACL (Zhang et al., 2023)	Wikipedia, 18 languages	7,611 (Test-A)	106M	binary	native-speaker annotators	yes
NTCIR-12 MathIR (Zanibbi et al., 2016)	arXiv, Wikipedia	107 topics	8.3M; 320K	graded	mathematics students	no
ARQMath-3 Task 1 (Mansouri et al., 2022b)	Math StackExchange	78	~1M questions	4 levels	paid mathematics and CS students	158 topics
BRIGHT math sets (Su et al., 2025)	olympiad, theorem QA	76–194	24K–188K	binary	source metadata (same technique or theorem); GPT-4 for TheoremQA-T	no
MathNet-Retrieve (Alshammari et al., 2026)	olympiad problems	15K	117K	equivalent	LLM-generated pairs kept when two LLM judges agree	no
SABER-Math (Georgiev et al., 2026)	school and olympiad solutions	1,000	283K	graded	pairwise LLM judge	no
MathPinpoint	mathematical web pages	6,173	3.5M	0/1/2	test: two LLM judges and an LLM arbiter; train: one LLM judge or a distilled 8B model	1.85M queries, 43.5M pairs

Table 1: MathPinpoint and related retrieval benchmarks. Figures are from each benchmark’s paper; ranges are over its datasets. MathPinpoint combines a same-problem, correctness-aware definition of relevance with a training split of 43.5M judged pairs.

Mathematical information retrieval. The NTCIR-12 MathIR task (Zanibbi et al., 2016) evaluated formula and keyword search over arXiv and Wikipedia with student assessors. ARQ-Math (Zanibbi et al., 2020; Mansouri et al., 2021, 2022a) evaluated answer retrieval and formula retrieval for Math Stack Exchange questions; each year, paid mathematics and computer-science students assessed 71 to 78 answer-retrieval topics on a four-level scale. Formula search engines such as Approach Zero (Zhong and Zanibbi, 2019; Zhong et al., 2020) and Tangent-CFT (Mansouri et al., 2019) target the structure of formulae. NaturalProofs (Welleck et al., 2021) retrieves the references a proof cites. MIRB (Ju and Dong, 2025) gathers twelve such datasets with their original labels. Recent benchmarks build their labels with LLMs: MathNet-Retrieve (Alshammari et al., 2026) generates an equivalent problem and hard negatives for each olympiad problem with an LLM and keeps pairs on which two LLM judges agree, SABER-Math (Georgiev et al., 2026) grades problem–solution pairs with a pairwise LLM judge, and Re²Math (Lyu et al., 2026) retrieves theorems for gaps in research-level proofs. Habibullah et al. (2026) show that a retriever trained on pairs written with MathNet-Retrieve’s own generation recipe gains up to 45 points of Recall@1 on its easiest tier, a warning that generated positives can reward matching the generator rather than the mathematics. MathPinpoint’s documents are real web pages, its queries are extracted from them (a page’s own question with minimal edits

where the page has one), and a query’s candidates are whatever retrieval returns from 3.5M pages.

Mathematical web corpora. OpenWebMath (Paster et al., 2024), MathPile (Wang et al., 2024), DeepSeekMath’s corpus (Shao et al., 2024), FineMath (Allal et al., 2025), MegaMath (Zhou et al., 2025) and UltraData-Math (Zhou et al., 2026; Wang et al., 2026) were built for language-model pre-training. MathPinpoint turns one of them into a retrieval collection with problem-level labels.

LLMs as relevance assessors. Thomas et al. (2024) found that LLM relevance labels agree with searcher preferences closely enough to be used in production, and UMBRELA (Upadhyay et al., 2024, 2025) reproduced the approach with open prompts on TREC collections; the LLM-Judge challenge (Rahmani et al., 2024) compared many such assessors. Faggioli et al. (2023), Clarke and Dietz (2025) and Soboroff (2025) argue that LLM labels cannot replace human assessment, in part because an LLM judge can share blind spots with the systems it evaluates. Distilling an LLM’s judgments into a smaller ranker is common practice (Sun et al., 2023), and automatic labels have been used to fill unjudged documents in existing collections (MacAvaney and Soldaini, 2023). MathPinpoint does not claim human-level labels: it records the judge of every training label, reports agreement between independent LLM judges, and measures the distilled 8B model on the pairs it labels.

Training data for dense retrieval. Dense retrievers (Karpukhin et al., 2020; Xiong et al., 2021; Reimers and Gurevych, 2019) learn from (query, positive, hard negative) triples, and synthetic queries generated from documents are a common source of them (Bonifacio et al., 2022; Dai et al., 2023). MathPinpoint’s extractor instead reuses a page’s own question whenever the page contains one and composes a query only when it does not, and its negatives are judged rather than assumed.

Incomplete judgments and pool bias. Pooled test collections judge only documents that some contributing system retrieved, so a new system can be penalised for relevant documents nobody judged (Zobel, 1998; Buckley et al., 2007). Measures such as bpref (Buckley and Voorhees, 2004), inferred AP and nDCG (Yilmaz and Aslam, 2006; Yilmaz et al., 2008) and condensed-list metrics (Sakai, 2007; Sakai and Kando, 2008), which skip unjudged documents, reduce this bias; AR-QMath’s primary measure, $nDCG'$, is such a condensed-list nDCG. BEIR measured the share of unjudged documents in each system’s top 10 on TREC-COVID and, after judging 980 of them, found that the $nDCG@10$ of a dense retriever rose from 0.654 to 0.735 (Thakur et al., 2021). We report standard $nDCG@10$ (Järvelin and Kekäläinen, 2002) together with the share of each system’s top 10 that is judged and condensed $nDCG@10$, and we measure the bias directly by judging the unjudged top-10 documents of two systems on a random sample of test queries.

3 The MathPinpoint Dataset

3.1 Desiderata

MathPinpoint is built on five desiderata. (a) **Problem-level relevance:** a document is relevant when it answers the problem that was asked, with its conditions and its target, not when it shares a topic. (b) **Correctness:** a document whose central mathematics is wrong is not relevant, however close its problem is. (c) **Scale:** the training split must be large enough to fine-tune modern embedding models, which rules out human assessment of every label. (d) **Provenance:** because the labels come from models of different strength, every label records which judge produced it, and the exact prompts that produced the data are released and pinned by SHA-256. (e) **Held-out evaluation:** the test queries share no problem with the training

queries, and their labels come from more than one judge.

3.2 Task and Rubric

Given a query, a self-contained mathematical question, a system ranks the documents of a fixed corpus. Every (query, document) pair is judged against one written rubric, *strict-math-retrieval-v1*, which scores usefulness toward a *correct* answer to *the same* problem:

- **2, highly relevant:** the document directly and substantially answers the same mathematical need, and its central reasoning and conclusion are correct. An equivalent method or notation is fine. For a query that asks only for a number or an expression, a correct short answer can score 2; when a proof or derivation is requested, an unsupported conclusion cannot.
- **1, partially relevant:** substantive, mathematically correct material that advances the same problem but leaves important work, such as a correct lemma, a method or a solved subcase. Topical overlap alone does not qualify.
- **0, not relevant or unusable:** a different problem, changed conditions, keyword overlap only, no answer, a major mathematical error, a correct-looking conclusion with no reliable basis, or mathematics that cannot be verified.

When the query itself cannot be interpreted, for example because it depends on a missing figure, the judge marks it unevaluable and gives no score. Each judgment is a JSON object with eight fields: the score, an *answer validity* (correct, minor error, major error, uncertain, not applicable), one or more of 18 *reason codes* such as *full_solution*, *condition_mismatch*, *target_mismatch*, *topic_only* and *major_math_error*, a confidence, a short rationale and identifiers. A JSON schema ties the reason codes and the validity to the score, so a 2 cannot carry *wrong_problem*, and a *major_error* cannot score above 0. The full list of codes is in Appendix A. Throughout the paper, a document is *relevant* when it scores 2; the test split’s relevance labels contain only such documents.

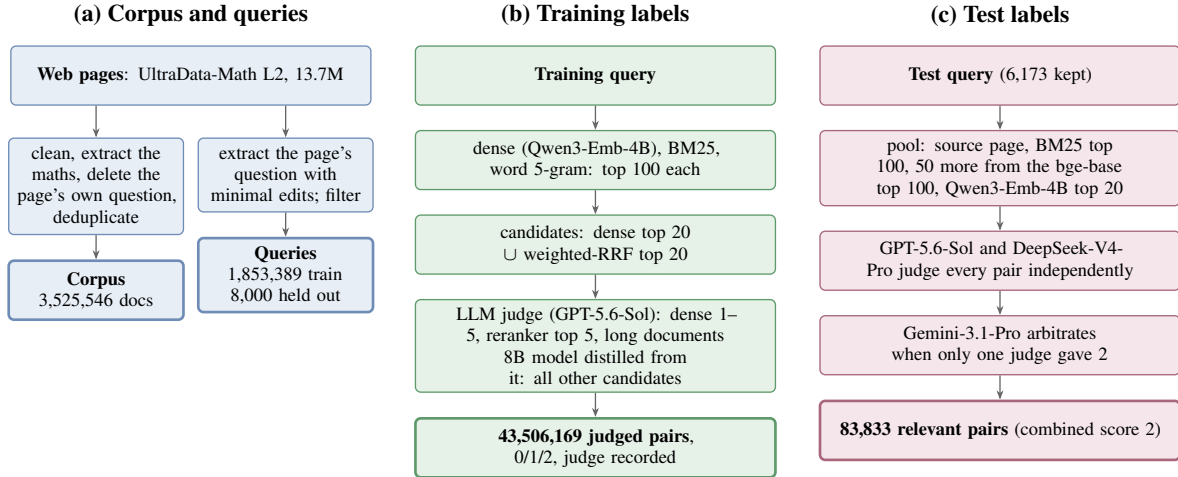


Figure 2: How MathPinpoint is built. (a) Web pages become the corpus and, through a minimal-edit extraction, the queries. (b) Each training query’s candidates come from three retrieval routes and are labeled by an LLM judge or by an 8B model distilled from it. (c) Each held-out test query’s pool is judged independently by two LLM judges, and a third arbitrates when they disagree about relevance.

3.3 Corpus

The pages come from the L2 tier of UltraData-Math (Zhou et al., 2026), web pages that passed a mathematical quality classifier. We used a copy of this tier that adds a 0–7 difficulty score to each page¹ and dropped the 1.4% of pages scored 0 or 1; exact deduplication of the raw text then removes 55.4% of the remaining 13.5M pages. An extraction model, Qwen3.5-2B (Qwen Team, 2026) distilled from GPT-5.6 and decoded greedily, turns each remaining page into its mathematical content.

Removing each page’s own question. Every query is extracted from a page (§3.4). If that page still contained the question word for word, retrieving it would be a string-matching exercise. The question span that the query extractor copied is therefore deleted from its page before indexing, with guards that never cut inside a formula and never delete more than 15% of a document or more than 500 characters. On 183,443 measured documents, the share that still contain their full question falls from 98.2% to 0.1%.

Deduplication and filtering. After Unicode normalisation, exact deduplication on a hash of the lowercased text removes 26.8% of the documents. Near-duplicates are found with MinHash (Broder, 1997) over word 5-grams and locality-sensitive hashing (Indyk and Motwani, 1998), keeping candidate pairs with an exact 5-gram Jaccard similar-

ity of at least 0.80. On a sample, 10.8% of the proposed deletions were wrong, so every deletion was reviewed: the 76.5% of proposals whose document has no content that the kept copy lacks are deleted, and the rest go to an LLM judge, which keeps 54.8% of those it reviews. Finally, 58,748 documents whose extraction degenerated into repetition and 93,917 documents that bundle five or more separate problems are removed. The corpus has **3,525,546** documents; Appendix B gives every step.

3.4 Queries

Queries are extracted from the pages, one per page, with a *minimal-edit* prompt. When a page contains a question somebody actually asked, that question is the query, copied with as few changes as possible: the extractor may only remove what is not part of the question (greetings, pleas for help, the asker’s failed attempt), insert a value or definition that the question refers to but that appears elsewhere on the same page, and replace a pointer such as “the equation above” with what it points to. It must not reword, normalise notation or translate. When a page has no question, as in an expository article, the extractor composes one. Of 5,929,019 documents, 85.1% yield a query, 64.8% of them taken from the page’s own question; the rest are declined, mostly because the page is a reference page, its text is corrupted, or it needs a missing figure. After the filtering below, 43.0% of training queries are a page’s own question. On a sample of 210,439 queries, the median share of a query’s

¹<https://huggingface.co/datasets/TeraflopAI/udml2-labeled>

characters that lie in a run of at least 15 characters copied from its page is 1.000. The extractor is GPT-5.6-Sol.

Filtering. Queries are deduplicated exactly after normalisation; near-duplicate merging was rejected because an LLM judge found 13–16% of pairs at word Jaccard 0.8–0.9 to be different problems. Queries that overlap with the 8,000 evaluation queries are removed, lexically (word 5-gram Jaccard ≥ 0.50 , checked exhaustively at the end) and semantically (embedding cosine ≥ 0.70 and judged the same problem by an LLM). Queries from pages that bundle five or more problems are removed.

Source-page check. Each query is then judged against its own source page by the LLM judge under the rubric. Only queries whose page scores 2 enter retrieval (70.0% of 2,901,088), so every query starts with at least one known full answer in the corpus. When near-duplicate removal took a query’s source page out, the surviving copy stands in for it if it is also a full answer.

Self-containedness. Retrievers see only the query, so it must make sense on its own. A second prompt asks whether a query depends on an undescribed figure, refers to values or options it does not state, depends on an external source, leaves its target unclear, or needs no mathematics. DeepSeek-V4-Pro (DeepSeek-AI et al., 2026) judges every query, and Gemini-3.1-Pro-Preview (Google DeepMind, 2026) re-judges the ones it flags; a query is removed when both find it not self-contained. On a 603-query calibration set the two judges agree on 87.7% of queries (Cohen’s $\kappa = 0.722$). This removes 7.8% of training queries; after a final exhaustive leakage check, **1,853,389** remain.

3.5 Candidates

Three routes each retrieve the top 100 documents for every training query: a dense retriever, Qwen3-Embedding-4B (Zhang et al., 2025), with documents embedded up to 32,768 tokens; BM25 (Robertson and Zaragoza, 2009) with $k_1 = 1.2$, $b = 0.75$ over lowercased alphanumeric tokens; and word 5-gram overlap. The routes are fused with weighted reciprocal rank fusion (Cormack et al., 2009), $\text{score}(d) = \sum_r w_r / (60 + \text{rank}_r(d))$, with weight 2 for the dense route and 1 for the others. A query’s candidates are its

dense top 20 and its fused top 20. The fused list adds 4.55 candidates per query on average, 10% of which only BM25 or 5-gram found. On a 1,000-query probe whose three-route pool was judged in full, the weighted fusion places 4.56 full answers per query in its top 20, against 3.99 for the dense route alone and 4.03 for fusion by best rank (Appendix C).

3.6 Judging the Training Candidates

Judging all 43.5M candidates with a frontier LLM would be expensive, so the LLM judge, GPT-5.6-Sol, is spent where it matters most, and an 8B relevance model distilled from it labels the rest (Table 2).

The LLM judge labels dense ranks 1–5, every document longer than 8,144 tokens among dense ranks 6–20 and the fused positions 1–5, and, within dense ranks 6–20, the five documents ranked highest by a zero-shot reranker. The reranker, Qwen3-Reranker-4B (Zhang et al., 2025), is told to judge whether a document answers “the same problem with the same particulars”. On a 196-query probe whose candidate pool was fully judged, it separates full answers from non-answers with an AUC of 0.921, against 0.599 for the fused retrieval order: retrieval is good at bringing relevant documents into the top 20 and poor at ordering them there. The LLM judge reads the first 30,000 characters of a document, or 100,000 for long documents.

The 8B relevance model is Qwen3-Reranker-8B with a new single-output regression head, trained with a mean-squared-error loss on 9.8M LLM labels of reranker picks and long documents from dense ranks 6–20, the same rank band as most pairs it labels. It reads the (query, document) pair up to 8,192 tokens. Its raw score is kept on every pair it scored, including the 8.7M pairs that also carry an LLM label; the released 0/1/2 label applies default cuts at 0.5 and 1.5, which users can replace. Section 4 measures it.

3.7 Test Split

The test split is built from 8,000 evaluation queries extracted by the same pipeline and held out from training. They pass the same self-containedness check (594 removed), and then a stricter one: DeepSeek-V4-Pro, Gemini-3.1-Pro-Preview and GPT-5.6-Sol each give one verdict, and a query

Candidates	Judge	Rows	0	1	2	unevaluable
Dense ranks 1–5	LLM	8,740,887	33.8%	31.6%	34.3%	0.30%
Dense ranks 6–20, reranker’s top 5	LLM (+8B score)	8,682,170	36.1%	39.1%	24.3%	0.51%
Dense ranks 6–20, longer than 8,144 tokens	LLM	1,275,579	56.4%	27.7%	15.1%	0.74%
Dense ranks 6–20, all other documents	8B	15,818,148	47.5%	44.2%	8.2%	–
Fused-only, long, fused positions 1–5	LLM	113,876	38.3%	33.3%	27.7%	0.63%
Fused-only, long, fused positions 6–20	8B	677,817	45.6%	38.2%	16.2%	–
Fused-only, up to 8,144 tokens	8B	7,609,881	37.2%	42.3%	20.5%	–
Query’s own source page	LLM	562,940	–	–	100%	–

Table 2: Who judges which training candidates, and the resulting labels. *Fused-only* candidates are in the fused top 20 but not in the dense top 20. A source page gets its own row only when no stage above covers it. Not shown: 24,871 rows in 8B stages that carry an LLM label. In total the LLM labels 19,400,323 rows and the 8B model 24,105,846.

The two judges	Combined	Pairs
agree	that score	77.2%
differ by one, neither gave 2	the lower score	13.7%
differ by two, or only one gave 2	arbiter decides	5.6%
either found the query unevaluable	pair dropped	3.4%

Table 3: Combining the two judges on the 947,537 first-round test pairs. The 0.06% of pairs whose arbitration returned no valid output take the lower score.

is removed when at least two find it not self-contained (241 more).

Pooling. Each test query’s pool contains its source page, the BM25 top 100, and the first 50 documents of the bge-base-en-v1.5 (Xiao et al., 2024b) top 100 not already pooled. The pools were retrieved over the documents from before near-duplicate removal, and only pooled documents that are in the corpus are kept: 947,537 (query, document) pairs. A second round adds the Qwen3-Embedding-4B top 20 over the corpus, of which 69,398 pairs were new.

Labels. Every pooled pair is judged independently by GPT-5.6-Sol and DeepSeek-V4-Pro under the same rubric, and the two scores are combined by fixed rules (Table 3). Gemini-3.1-Pro-Preview arbitrates only disagreements that cross the line between relevant and not relevant. A document is relevant when its combined score is 2. After both self-containedness checks, the 6,173 queries with at least one relevant document form the test split, with **83,833** relevant pairs. Documents outside a query’s pool were never judged and count as not relevant (§6).

	Train	Test
Queries	1,853,389	6,173
Query length, median	164	165
Judged pairs	43,506,169	794,380 [†]
labeled 2	8,888,149	83,833
labeled 1	17,025,106	159,033
labeled 0	17,512,014	551,514
query unevaluable	80,900	dropped
Relevant docs per query	–	13.6
Corpus documents	3,525,546	
Document length, median	3,957	

Table 4: MathPinpoint at a glance; lengths in characters. [†]Combined labels of the pooled pairs of the 6,173 test queries; pairs whose query a judge found unevaluable are dropped, and the release contains only the relevant pairs. Appendix E gives the full descriptive statistics.

3.8 Statistics and Format

Table 4 summarises the release. Documents are long, with a median of 3,957 characters, and queries short, with a median of 164. A training query has 23.5 judged candidates on average. A test query has a median of 4 relevant documents but a mean of 13.6 and a maximum of 140: standard textbook problems are answered by many pages. The release is a Hugging Face dataset with four configurations: corpus, queries (train and test), judgments (train) and qrels (test). Identifiers are content hashes, so anyone holding the text can recompute them.

4 Label Quality

No label in MathPinpoint was assigned by a person. We therefore measure each judge against independent judgments by other LLMs. Agreement between two LLMs is not accuracy, since both can be wrong in the same way, but it shows how much a label depends on the choice of judge and where

	LLM judge	Second judge	Combined
score 0	55.0%	58.2%	68.6%
score 1	35.4%	28.4%	22.8%
score 2	9.6%	13.4%	8.5%

Table 5: Label distribution of the LLM judge (GPT-5.6-Sol), an independent second judge (DeepSeek-V4-Pro) and their combination on 306,993 training candidates.

the judges disagree.

The LLM judge against a second judge.

DeepSeek-V4-Pro independently re-judged 337,032 training candidates under the same rubric, and the pairs were combined with the test-split rules (Table 3); 306,993 have a combined label. The two judges agree exactly on 77.6% of pairs, 7.2% needed arbitration, and only 1.0% are a 0 against a 2: the judges disagree about how strict to be, not about which problem is asked. GPT-5.6-Sol is the more lenient at the 0/1 boundary. It calls 45.0% of candidates at least partially relevant, against 31.4% for the combined label, while its share of 2s (9.6%) is close to the combined share (8.5%; Table 5). Score 1 is the least stable label and is best read as “partially relevant, uncertain”.

Source pages. Every training query enters retrieval only if its own source page was judged a full answer. DeepSeek-V4-Pro re-judged 1,453,403 of these 2,032,033 positives and gave 95.0% of them a 2, 4.1% a 1 and 0.85% a 0. Because the second judge gives 2 more readily than the first on retrieval candidates (13.4% against 9.6%), 95% agreement is an upper bound and 0.85% strong disagreement a lower bound. When a source page is also among the candidates the LLM judges, it is judged a second time and the table keeps the second label, which is not always 2: 49,024 of the 1,788,845 pairs marked as a query’s source page are not, and 19,797 queries (1.1%) have no row labeled 2.

The 8B model. Most of the variance of the 0/1/2 labels, 61.2% on a held-out set of 31,830 pairs over 6,000 queries, lies between queries, which makes pooled R^2 flattering. We therefore report within-query R^2 , the share of within-query variance the model explains. At the 10,000-step checkpoint that labels the data it is 0.364 [0.349, 0.380]; a 0.6B model trained the same way reaches only 0.140, so capacity matters. With the default cuts, the 8B label matches the LLM label on 73.2% of

	Dense 6–20	Fused short	Fused long	All three
Exact agreement	78.1	76.3	79.2	77.6
LLM 2 kept as 2	56.6	73.6	70.6	65.0
0 $\leftrightarrow$ 2 flips	0.41	0.75	0.56	0.52
8B 2 that is an LLM 2	83.2	83.7	83.1	83.5
8B 0 that is an LLM 2	0.5	0.5	0.5	0.5

Table 6: The 8B model against the LLM judge on 3,600 sampled pairs from the stages the 8B labels on its own (percentages; cuts at 0.5 and 1.5). The 23 pairs the LLM judged unevaluable are left out.

held-out pairs. The regression pulls toward the middle: a third of the LLM’s 2s receive an 8B label of 1.

The held-out set consists of reranker picks and long documents, not of the candidates the 8B model labels on its own. To measure it there, the LLM judge judged a stratified sample of 3,600 pairs from the three 8B stages, 400 per stage and 8B label, weighted back to the stages’ sizes (Table 6). Training uses an 8B 2 as a positive and an 8B 0 as a negative, so the two numbers that matter most are the last two rows: 83.5% of 8B positives are LLM positives, and most of the rest are partial answers, while only 0.5% of 8B negatives are LLM positives.

Test labels. On the 947,537 first-round test pairs the two judges agree on 77.2% (Table 3), about the same rate as on training candidates. Arbitration decides 5.6% of pairs, and 3.4% are dropped because one judge found the query unevaluable. In the second pooling round the rates are similar: of 69,398 new pairs, 71.3% agree, 16.9% take the lower score, 10.8% are arbitrated and 0.9% are dropped; the 28 pairs whose arbitration failed take the lower score.

5 Experiments

5.1 Evaluation Protocol

A system ranks all 3,525,546 corpus documents for each of the 6,173 test queries. The relevant documents of a query are its *qrels*, the documents with a combined score of 2. We report **nDCG@10** (Järvelin and Kekäläinen, 2002) and **Recall@100**, the share of a query’s relevant documents found in the top 100; both average over queries and count documents that were never judged as not relevant. Because every relevant document has the same grade, graded and binary nDCG coincide. To show how much

of each ranking the labels cover, we also report **judged@10**, the share of a system’s top 10 that has a combined label for that query, and **condensed nDCG@10** (Sakai, 2007; Sakai and Kando, 2008), nDCG@10 computed after removing unjudged documents from the ranking; ARQ-Math’s primary metric, nDCG’, is the same measure at a deeper cutoff. Intervals are 95% paired bootstrap intervals over queries with 10,000 resamples (Sakai, 2006). Code for the two standard metrics, a dozen lines of Python, is given in the dataset card.

5.2 Baselines

We evaluate one lexical and five dense retrievers (Table 7). **BM25** uses the same formulation as the candidate route ($k_1 = 1.2$, $b = 0.75$, lowercased alphanumeric tokens), with document statistics from the released corpus. **bge-base-en-v1.5** (Xiao et al., 2024b) uses CLS pooling and its recommended query instruction. The **Qwen3-Embedding** models (Zhang et al., 2025) use last-token pooling and the instruction “*Given a web search query, retrieve relevant passages that answer the query*” on queries only. Every dense model reads documents up to 512 tokens, embeddings are L2-normalised, and search is an exact inner product over the whole corpus.

Three of these systems also contributed to the test pool, in slightly different configurations: the BM25 top 100 and up to 50 further documents from the bge-base top 100, both retrieved over the corpus before near-duplicate removal, and the Qwen3-Embedding-4B top 20 with documents up to 32,768 tokens and a mathematics-specific instruction. Pool membership raises judged@10 and, with it, standard nDCG@10, so the table marks it.

BM25 is a strong baseline. On the full test split it has the highest standard nDCG@10, 0.562, and Recall@100, 0.738, but its own top 100 was pooled, so 99.4% of its top 10 is judged, while the dense retrievers are penalised for relevant documents they find outside the pool. On the 500-query sample on which the top 10 of BM25, bge-base and both 0.6B models is judged almost completely, BM25 (0.562) and the fine-tuned model (0.577) are tied: the difference is +0.015 [−0.025, +0.054]. Both are well ahead of the base model (0.465; the fine-tuned model leads it by 0.111 [0.081, 0.140]) and of bge-base (0.376). We read BM25’s strength as a property of the task: the same problem re-

curs almost verbatim across pages, and the numbers and variable names it shares are exactly what BM25 matches, whereas a dense retriever has to learn that a page with one number changed is a different problem (Figure 1). [TBD: one sentence on Qwen3-Embedding-4B and 8B]

5.3 Fine-Tuning on MathPinpoint

To test whether the training split teaches a retriever something the test split can measure, we fine-tune Qwen3-Embedding-0.6B and compare it with the same model before fine-tuning. The training recipe and the comparison were registered before the final training runs.

Training rows. Each training query gives one row. Its positive is drawn at random from its candidates labeled 2, and its three hard negatives are the candidates labeled 0 with the best dense rank; queries with fewer than three such negatives are skipped. Where both judges labeled a pair the LLM label is used, and a pair labeled only by the 8B model counts as positive when its raw score is at least 1.5 and as negative when it is below 0.5. Rows from the source-page check are not used, but a source page that retrieval also returned is a candidate like any other; it is the positive in 46.8% of rows. Documents are cut to 4,000 characters and queries to 2,000, which gives 1,225,057 rows.

Setup. We train with the multiple-negatives ranking loss (Henderson et al., 2017), using in-batch negatives and the three hard negatives, combined with a Matryoshka loss (Kusupati et al., 2022) over 768, 512, 256 and 128 dimensions. The effective batch is 512 (8 GPUs $\times$ 64, with GradCache (Gao et al., 2021) at mini-batch 32), the learning rate 2×10^{-5} with 10% warm-up, the sequence length 512, and training runs for 1,000 steps, so a run sees 512,000 of the 1,225,057 rows. We train three runs with seeds 42, 43 and 44 and average their per-query scores. Evaluation follows §5.1.

Results. Fine-tuning raises nDCG@10 from 0.441 to 0.477, a difference of +0.036 [+0.028, +0.045] (Table 8); each of the three seeds lands between 0.476 and 0.478. Recall@100 barely moves (+0.007 [−0.001, +0.016]). These differences are measured against labels that cover the two models unequally, which Section 6 examines: corrected for coverage, the nDCG@10 gain is between +0.11 and +0.13.

System	In pool	All 6,173 queries		Pool coverage		500, judged
		nDCG@10	R@100	judged@10	cond.	nDCG@10
BM25	✓	0.562	0.738	99.4%	0.563	0.562
bge-base-en-v1.5	✓	0.377	0.505	99.5%	0.378	0.376
Qwen3-Embedding-0.6B		0.441	0.561	66.4%	0.486	0.465
Qwen3-Embedding-4B	(✓)	?	?	?	?	—
Qwen3-Embedding-8B		?	?	?	?	—
Qwen3-Embedding-0.6B, fine-tuned		0.477	0.568	35.0%	0.617	0.577

Table 7: Retrievers on the MathPinpoint test split. *In pool*: the system’s ranking contributed documents to the judged pool; (✓) the same model contributed in a different configuration. *cond.*: condensed nDCG@10. The last column (500, judged) is nDCG@10 on a random sample of 500 test queries on which the top 10 of each listed system is at least 98.9% judged, because the unjudged top-10 documents of both Qwen3-Embedding-0.6B models were judged for this sample (§6.1); the two larger models’ gaps were not judged. The fine-tuned model is the mean of three seeds (§5.3).

	Before	After	Difference [95% CI]
nDCG@10	.441	.477	+0.036 [+0.028, +0.045]
R@100	.561	.568	+0.007 [−.001, +0.016]
cond. nDCG@10	.486	.617	+0.131 [+0.123, +0.139]
judged@10	66.4%	35.0%	

Table 8: Qwen3-Embedding-0.6B before and after fine-tuning on MathPinpoint (test split, 6,173 queries; after: mean of three seeds).

6 Analysis

6.1 Pool Bias

Standard nDCG@10 treats every unjudged document as not relevant. The fine-tuned model ranks many documents that none of the pooled systems retrieved: only 35% of its top 10 is judged, against 66% for the model before fine-tuning (Table 8). Its relevant documents outside the pool are invisible to the metric, so the standard difference understates the gain. Two independent corrections agree that the understatement is large.

Condensed lists. Removing unjudged documents before taking the top 10 gives 0.486 before and 0.617 after fine-tuning, a difference of +0.131 [+0.123, +0.139], more than three times the standard +0.036.

Judging the gap. On a random sample of 500 test queries we judged every unjudged document in both models’ top 10 under the test protocol, 6,810 pairs in all. With these labels, judged@10 on the sample rises from 67.3% to 99.3% for the model before fine-tuning and from 35.1% to 99.1% for the fine-tuned model, and the nDCG@10 difference grows from +0.043 [+0.013, +0.074] to +0.111 [+0.081, +0.140], close to the

condensed estimate.

Pool composition. Which systems contribute to the pool also shifts comparisons. The second pooling round, which added the Qwen3-Embedding-4B top 20, raised judged@10 from 58.0% to 66.4% for the model before fine-tuning but only from 32.0% to 35.0% for the fine-tuned model, and it moved the measured nDCG@10 difference from +0.042 to +0.036 and the Recall@100 difference from +0.016 to +0.007. A pooled system that resembles one of the compared systems, here a larger model of the same family as the base model, favours that system.

Across retrievers. Pool membership decides much of the standard ranking in Table 7. On the full test split, standard nDCG@10 puts BM25 0.085 ahead of the fine-tuned model; on the 500 queries where both rankings are judged almost in full, the two are tied (+0.015 [−0.025, +0.054] for the fine-tuned model). The standard gap is pool bias: BM25’s top 100 is in the pool, and most of the fine-tuned model’s top 10 is not. Condensed nDCG@10 errs in the other direction, putting the fine-tuned model 0.054 ahead, but it is closer to the fully judged comparison than the standard measure is. [TBD: Qwen3-Embedding-4B and 8B]

Recommendations. A retriever that differs from BM25, bge-base and Qwen3-Embedding-4B will find relevant documents that nobody judged, and BM25, whose top 100 is fully pooled, is the system that standard metrics favour most. When comparing systems on the test split we recommend reporting judged@10 next to nDCG@10, preferring comparisons between systems with similar coverage, and reading condensed nDCG@10 as a check

on the direction of a difference. For a definitive comparison, the unjudged top-10 documents of the compared systems can be judged with the released prompts, as we did for the sample above.

6.2 Many Answers per Problem

Some queries have many full answers. In a deep-pool probe of 1,000 queries with 725,326 pairs judged by the LLM judge, the number of full answers per query has a median of 7, a 90th percentile of 230 and a maximum of 1,840: standard textbook problems are answered by hundreds of pages. For such a query, most of its full answers lie outside the 20 to 40 candidates that the training split labels, and an unlabeled training candidate is not evidence of irrelevance. The same holds for Recall@100 on the test split, which is recall over the judged relevant documents only.

6.3 Why Judge Retrieved Candidates

Every training query comes with a known full answer, its source page, so one could train on (query, source page) pairs alone. That teaches a retriever to find the page a query was copied from rather than every page that answers it. In an earlier experiment on a previous test set of 2,580 queries, fine-tuning bge-base-en-v1.5 on 300,000 (query, source page) pairs left nDCG@10 unchanged (+0.007 [−0.003, +0.017]) and cut Recall@100 by 0.088 [0.078, 0.096] relative to the untrained model. MathPinpoint therefore labels the retrieved candidates as well; in the training rows of §5.3, the source page is the positive in only 46.8% of rows.

7 Conclusion

MathPinpoint frames math retrieval as finding the page that solves a given problem correctly, and provides the data to train and test retrievers for it: a corpus of 3.5M mathematical web pages, 1.85M training queries with 43.5M judged candidates whose judge is recorded on every row, and 6,173 test queries labeled by two LLM judges and an arbiter. Fine-tuning a strong embedding model on the training split improves it on the test split, by more than the standard metric shows, and brings it level with BM25, which is a strong baseline for a task in which the same problem recurs almost verbatim across the web. The same analysis carries a general lesson for benchmarks built from pooled LLM judgments: the share of a system’s ranking that the labels cover should be reported with every

score, because at this scale it decides which system looks better. We release the data, the exact prompts that produced it and the evaluation code, so that the labels can be audited, extended to new systems’ rankings, and replaced by better judges.

Limitations

Coverage. Only each training query’s dense top 20 and fused top 20 are labeled, so an unlabeled document is not evidence of irrelevance. Some queries have hundreds of full answers in the corpus (§6.2), most of which are not among their labeled candidates.

Label quality. The labels are not human gold, and no person reviewed them. The rubric’s text describes qualified human review for some cases, such as disagreements between a 0 and a 2; that review was not carried out for this release. Each training label comes from a single LLM judge or from the 8B model distilled from it; test labels combine two LLM judges and an arbiter. The LLM judge is lenient at the boundary between 0 and 1, and score 1 is the least stable label. The 8B scores regress toward the middle: a third of the LLM’s 2s receive an 8B label of 1, and the cuts at 0.5 and 1.5 are defaults rather than calibrated thresholds, so the raw score ships with every 8B label. About one in six 8B positives is a partial answer for the LLM judge (§4).

Corpus. Near-duplicate pairs just above the deduplication threshold can survive. A query’s question is removed from its own page but can still appear on other pages that repost the same problem.

Test split. Only pooled documents were judged, and the pools come from BM25, bge-base-en-v1.5 and Qwen3-Embedding-4B. A retriever that ranks differently retrieves documents that nobody judged: nDCG@10 counts them as not relevant, and Recall@100 is recall over the judged relevant documents only. Pool bias can therefore move either metric, and the difference between two retrievers, in either direction (§6.1).

Ethics Statement

Data and licenses. The documents are cleaned extractions of public web pages distributed in UltraData-Math under Apache-2.0. We release the corpus, queries and labels under CC0 1.0 and the prompts, schemas and code under Apache-2.0;

rights in the original web pages remain with their owners. The pages carry no URLs. We did not run a dedicated filter for personal information; the pages come from a tier selected by a mathematical quality classifier.

Model-generated labels. Every label was produced by a language model, and the LLM judges and the 8B model can share systematic errors that agreement between them does not reveal. We release each row’s judge and the 8B model’s raw score so that users can choose which labels to trust, and the exact prompts so that the labels can be audited or regenerated.

Use. A retriever that finds worked solutions to a given problem can help learners check their work and can also be used to look up homework answers; this is true of any search engine over the same pages. We release the data to improve retrieval and its evaluation.

References

- Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, and 3 others. 2025. [SmolLM2: When smol goes big – data-centric training of a small language model](#). Preprint, arXiv:2502.02737.
- Shaden Alshammari, Kevin Wen, Abrar Zainal, Mark Hamilton, Navid Safaei, Sultan Albarakati, William T. Freeman, and Antonio Torralba. 2026. [MathNet: A global multimodal benchmark for mathematical reasoning and retrieval](#). In *The Fourteenth International Conference on Learning Representations, ICLR 2026, Rio de Janeiro, Brazil, April 23-27, 2026*, pages 12477–12507.
- Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. 2018. [MS MARCO: A human generated MACHine reading COMprehension dataset](#). Preprint, arXiv:1611.09268.
- Luiz Henrique Bonifacio, Hugo Queiroz Abonizio, Marzieh Fadaee, and Rodrigo Nogueira. 2022. [InPars: Unsupervised dataset generation for information retrieval](#). In *SIGIR ’22: The 45th International ACM SIGIR Conference on Research and Development in Information Retrieval, Madrid, Spain, July 11-15, 2022*, pages 2387–2392. ACM.
- Andrei Z. Broder. 1997. [On the resemblance and containment of documents](#). In *Compression and Complexity of SEQUENCES 1997, Positano, Amalfitan Coast, Salerno, Italy, June 11-13, 1997, Proceedings*, pages 21–29. IEEE.
- Chris Buckley, Darrin Dimmick, Ian Soboroff, and Ellen M. Voorhees. 2007. [Bias and the limits of pooling for large collections](#). *Information Retrieval*, 10(6):491–508.
- Chris Buckley and Ellen M. Voorhees. 2004. [Retrieval evaluation with incomplete information](#). In *SIGIR 2004: Proceedings of the 27th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Sheffield, UK, July 25-29, 2004*, pages 25–32. ACM.
- Charles L. A. Clarke and Laura Dietz. 2025. [LLM-based relevance assessment still can’t replace human relevance assessment](#). In *Proceedings of the Eleventh International Workshop on Evaluating Information Access, EVIA 2025, a Satellite Workshop of the NTCIR-18 Conference, Tokyo, Japan, June 10, 2025*. National Institute of Informatics (NII).
- Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. 2009. [Reciprocal rank fusion outperforms condorcet and individual rank learning methods](#). In *Proceedings of the 32nd Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2009, Boston, MA, USA, July 19-23, 2009*, pages 758–759. ACM.
- Zhuyun Dai, Vincent Y. Zhao, Ji Ma, Yi Luan, Jianmo Ni, Jing Lu, Anton Bakalov, Kelvin Guu, Keith B. Hall, and Ming-Wei Chang. 2023. [Promptagator: Few-shot dense retrieval from 8 examples](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Tri Dao. 2024. [FlashAttention-2: Faster attention with better parallelism and work partitioning](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- DeepSeek-AI et al. 2026. [DeepSeek-V4: Towards highly efficient million-token context intelligence](#). Preprint, arXiv:2606.19348.
- Kenneth Enevoldsen, Isaac Chung, Imene Kerboua, Márton Kardos, Ashwin Mathur, David Stap, Jay Gala, Wissam Siblini, Dominik Krzemiński, Genta Indra Winata, Saba Sturua, Saiteja Utpala, Mathieu Ciancone, Marion Schaeffer, Diganta Misra, Shreeya Dhakal, Jonathan Rystrom, Roman Solomatin, Ömer Çağatan, et al. 2025. [MMTEB: Massive multilingual text embedding benchmark](#). In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Guglielmo Faggioli, Laura Dietz, Charles L. A. Clarke, Gianluca Demartini, Matthias Hagen, Claudia Hauff,

- Noriko Kando, Evangelos Kanoulas, Martin Potthast, Benno Stein, and Henning Wachsmuth. 2023. [Perspectives on large language models for relevance judgment](#). In *Proceedings of the 2023 ACM SIGIR International Conference on Theory of Information Retrieval, ICTIR 2023, Taipei, Taiwan, 23 July 2023*, pages 39–50. ACM.
- Luyu Gao, Yunyi Zhang, Jiawei Han, and Jamie Callan. 2021. [Scaling deep contrastive learning batch size under memory limited setup](#). In *Proceedings of the 6th Workshop on Representation Learning for NLP (RepL4NLP-2021)*, pages 316–321, Online. Association for Computational Linguistics.
- Nikolay Georgiev, Maria Drencheva, Kseniia Ibragimova, Ivo Petrov, Dimitar I. Dimitrov, and Martin Vechev. 2026. [SABER-Math: Automated benchmark for information retrieval evaluation in mathematics](#). *Preprint*, arXiv:2606.29894.
- Google DeepMind. 2026. [Gemini 3.1 Pro model card](#). <https://deepmind.google/models/model-cards/gemini-3-1-pro/>. Published February 2026.
- Ali Habibullah, Mohammad Alshiekh, Yazan Alshoibi, Salman Khan, and Naeemullah Khan. 2026. [Recipe-matching, not equivalence](#). *Preprint*, arXiv:2609.31927.
- Matthew Henderson, Rami Al-Rfou, Brian Strope, Yunhsuan Sung, Laszlo Lukacs, Ruiqi Guo, Sanjiv Kumar, Balint Miklos, and Ray Kurzweil. 2017. [Efficient natural language response suggestion for Smart Reply](#). *Preprint*, arXiv:1705.00652.
- Piotr Indyk and Rajeev Motwani. 1998. [Approximate nearest neighbors: Towards removing the curse of dimensionality](#). In *Proceedings of the Thirtieth Annual ACM Symposium on the Theory of Computing, Dallas, Texas, USA, May 23-26, 1998*, pages 604–613. ACM.
- Kalervo Järvelin and Jaana Kekäläinen. 2002. [Cumulated gain-based evaluation of IR techniques](#). *ACM Transactions on Information Systems*, 20(4):422–446.
- Haocheng Ju and Bin Dong. 2025. [MIRB: Mathematical information retrieval benchmark](#). *Preprint*, arXiv:2505.15585.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. [Dense passage retrieval for open-domain question answering](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online. Association for Computational Linguistics.
- Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham M. Kakade, Prateek Jain, and Ali Farhadi. 2022. [Matryoshka representation learning](#). In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, pages 30233–30249. Curran Associates, Inc.
- Zicheng Lyu, Wenjie Yang, Shengzhong Zhang, and Zengfeng Huang. 2026. [Re²Math: Benchmarking theorem retrieval in research-level mathematics](#). *Preprint*, arXiv:2605.09012.
- Sean MacAvaney and Luca Soldaini. 2023. [One-shot labeling for automatic relevance estimation](#). In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2023, Taipei, Taiwan, July 23-27, 2023*, pages 2230–2235. ACM.
- Behrooz Mansouri, Vít Novotný, Anurag Agarwal, Douglas W. Oard, and Richard Zanibbi. 2022a. [Overview of ARQMath-3 \(2022\): Third CLEF lab on answer retrieval for questions on math](#). In *Experimental IR Meets Multilinguality, Multimodality, and Interaction - 13th International Conference of the CLEF Association, CLEF 2022, Bologna, Italy, September 5-8, 2022, Proceedings*, volume 13390 of *Lecture Notes in Computer Science*, pages 286–310. Springer.
- Behrooz Mansouri, Vít Novotný, Anurag Agarwal, Douglas W. Oard, and Richard Zanibbi. 2022b. [Overview of ARQMath-3 \(2022\): Third CLEF lab on answer retrieval for questions on math \(working notes version\)](#). In *Proceedings of the Working Notes of CLEF 2022 - Conference and Labs of the Evaluation Forum, Bologna, Italy, September 5th - to - 8th, 2022*, volume 3180 of *CEUR Workshop Proceedings*, pages 1–27. CEUR-WS.org.
- Behrooz Mansouri, Shaurya Rohatgi, Douglas W. Oard, Jian Wu, C. Lee Giles, and Richard Zanibbi. 2019. [Tangent-CFT: An embedding model for mathematical formulas](#). In *Proceedings of the 2019 ACM SIGIR International Conference on Theory of Information Retrieval, ICTIR 2019, Santa Clara, CA, USA, October 2-5, 2019*, pages 11–18. ACM.
- Behrooz Mansouri, Richard Zanibbi, Douglas W. Oard, and Anurag Agarwal. 2021. [Overview of ARQMath-2 \(2021\): Second CLEF lab on answer retrieval for questions on math](#). In *Experimental IR Meets Multilinguality, Multimodality, and Interaction - 12th International Conference of the CLEF Association, CLEF 2021, Virtual Event, September 21-24, 2021, Proceedings*, volume 12880 of *Lecture Notes in Computer Science*, pages 215–238. Springer.
- Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. 2023. [MTEB: Massive text embedding benchmark](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037, Dubrovnik, Croatia. Association for Computational Linguistics.

- OpenAI. 2026. [GPT-5.6 system card](https://deploymentsafety.openai.com/gpt-5-6). OpenAI Deployment Safety Hub, <https://deploymentsafety.openai.com/gpt-5-6>. Dated 2026-07-09.
- Keiran Paster, Marco Dos Santos, Zhangir Azerbayev, and Jimmy Ba. 2024. [OpenWebMath: An open dataset of high-quality mathematical web text](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Qwen Team. 2026. [Qwen3.5: Towards native multimodal agents](#). Qwen blog, <https://qwen.ai/blog?id=qwen3.5>.
- Hossein A. Rahmani, Emine Yilmaz, Nick Craswell, Bhaskar Mitra, Paul Thomas, Charles L. A. Clarke, Mohammad Aliannejadi, Clemencia Siro, and Guglielmo Faggioli. 2024. [LLMJudge: LLMs for relevance judgments](#). In *Proceedings of The First Workshop on Large Language Models for Evaluation in Information Retrieval (LLM4Eval 2024) co-located with 10th International Conference on Online Publishing (SIGIR 2024), Washington D.C., USA, July 18, 2024*, volume 3752 of *CEUR Workshop Proceedings*, pages 1–3. CEUR-WS.org.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-BERT: Sentence embeddings using Siamese BERT-networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.
- Stephen E. Robertson and Hugo Zaragoza. 2009. [The probabilistic relevance framework: BM25 and beyond](#). *Foundations and Trends in Information Retrieval*, 3(4):333–389.
- Tetsuya Sakai. 2006. [Evaluating evaluation metrics based on the bootstrap](#). In *SIGIR 2006: Proceedings of the 29th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Seattle, Washington, USA, August 6-11, 2006*, pages 525–532. ACM.
- Tetsuya Sakai. 2007. [Alternatives to Bpref](#). In *SIGIR 2007: Proceedings of the 30th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Amsterdam, The Netherlands, July 23-27, 2007*, pages 71–78. ACM.
- Tetsuya Sakai and Noriko Kando. 2008. [On information retrieval metrics designed for evaluation with incomplete relevance assessments](#). *Information Retrieval*, 11(5):447–470.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. [DeepSeekMath: Pushing the limits of mathematical reasoning in open language models](#). *Preprint*, arXiv:2402.03300.
- Ian Soboroff. 2025. [Don’t use LLMs to make relevance judgments](#). *Information Retrieval Research*, 1(1):29–46.
- Hongjin Su, Howard Yen, Mengzhou Xia, Weijia Shi, Niklas Muennighoff, Han-yu Wang, Haisu Liu, Quan Shi, Zachary S. Siegel, Michael Tang, Ruoxi Sun, Jinsung Yoon, Sercan Ö. Arik, Danqi Chen, and Tao Yu. 2025. [BRIGHT: A realistic and challenging benchmark for reasoning-intensive retrieval](#). In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. 2023. [Is ChatGPT good at search? investigating large language models as re-ranking agents](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14918–14937, Singapore. Association for Computational Linguistics.
- Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. [BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models](#). In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks 1, NeurIPS Datasets and Benchmarks 2021, December 2021, virtual*.
- Paul Thomas, Seth Spielman, Nick Craswell, and Bhaskar Mitra. 2024. [Large language models can accurately predict searcher preferences](#). In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, pages 1930–1940. ACM.
- Shivani Upadhyay, Ronak Pradeep, Nandan Thakur, Daniel Campos, Nick Craswell, Ian Soboroff, and Jimmy Lin. 2025. [A large-scale study of relevance assessments with large language models using UMBRELA](#). In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval, ICTIR 2025, Padua, Italy, 18 July 2025*, pages 358–368. ACM.
- Shivani Upadhyay, Ronak Pradeep, Nandan Thakur, Nick Craswell, and Jimmy Lin. 2024. [UMBRELA: Umbrela is the \(open-source reproduction of the\) Bing RElevance assessor](#). *Preprint*, arXiv:2406.06519.
- Yudong Wang, Zixuan Fu, Hengyu Zhao, Chen Zhao, Chuyue Zhou, Xinle Lin, Hongya Lyu, Shuai Kang Xue, Yi Yi, Yingjiao Wang, Zhi Zheng, Yuzhou Zhang, Jie Zhou, Chaojun Xiao, Xu Han, Zhiyuan Liu, and Maosong Sun. 2026. [Data science and technology towards AGI part I: Tiered data management](#). *Preprint*, arXiv:2602.09003.
- Zengzhi Wang, Xuefeng Li, Rui Xia, and Pengfei Liu. 2024. [MathPile: A billion-token-scale pretraining](#).

- corpus for math. In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10-15, 2024*, pages 25426–25468. Curran Associates, Inc.
- Sean Welleck, Jiacheng Liu, Ronan Le Bras, Hanna Hajishirzi, Yejin Choi, and Kyunghyun Cho. 2021. **NaturalProofs: Mathematical theorem proving in natural language**. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks 1, NeurIPS Datasets and Benchmarks 2021, December 2021, virtual*.
- Chenghao Xiao, G. Thomas Hudson, and Noura Al Moubayed. 2024a. **RAR-b: Reasoning as retrieval benchmark**. *Preprint*, arXiv:2404.06347.
- Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muenighoff, Defu Lian, and Jian-Yun Nie. 2024b. **C-Pack: Packed resources for general Chinese embeddings**. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, pages 641–649. ACM.
- Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul N. Bennett, Junaid Ahmed, and Arnold Overwijk. 2021. **Approximate nearest neighbor negative contrastive learning for dense text retrieval**. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net.
- Emine Yilmaz and Javed A. Aslam. 2006. **Estimating average precision with incomplete and imperfect judgments**. In *Proceedings of the 2006 ACM CIKM International Conference on Information and Knowledge Management, Arlington, Virginia, USA, November 6-11, 2006*, pages 102–111. ACM.
- Emine Yilmaz, Evangelos Kanoulas, and Javed A. Aslam. 2008. **A simple and efficient sampling method for estimating AP and NDCG**. In *Proceedings of the 31st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2008, Singapore, July 20-24, 2008*, pages 603–610. ACM.
- Richard Zanibbi, Akiko Aizawa, Michael Kohlhase, Iadh Ounis, Goran Topić, and Kenny Davila. 2016. **NTCIR-12 MathIR task overview**. In *Proceedings of the 12th NTCIR Conference on Evaluation of Information Access Technologies, National Center of Sciences, Tokyo, Japan, June 7-10, 2016*, pages 299–308. National Institute of Informatics (NII).
- Richard Zanibbi, Douglas W. Oard, Anurag Agarwal, and Behrooz Mansouri. 2020. **Overview of AR-QMath 2020: CLEF lab on answer retrieval for questions on math**. In *Experimental IR Meets Multilinguality, Multimodality, and Interaction - 11th International Conference of the CLEF Association, CLEF 2020, Thessaloniki, Greece, September 22-25, 2020, Proceedings*, volume 12260 of *Lecture Notes in Computer Science*, pages 169–193. Springer.
- Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamaloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023. **MIRACL: A multilingual retrieval dataset covering 18 diverse languages**. *Transactions of the Association for Computational Linguistics*, 11:1114–1131.
- Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. **Qwen3 embedding: Advancing text embedding and reranking through foundation models**. *Preprint*, arXiv:2506.05176.
- Wei Zhong, Shaurya Rohatgi, Jian Wu, C. Lee Giles, and Richard Zanibbi. 2020. **Accelerating substructure similarity search for formula retrieval**. In *Advances in Information Retrieval - 42nd European Conference on IR Research, ECIR 2020, Lisbon, Portugal, April 14-17, 2020, Proceedings, Part I*, volume 12035 of *Lecture Notes in Computer Science*, pages 714–727. Springer.
- Wei Zhong and Richard Zanibbi. 2019. **Structural similarity search for formulas using leaf-root paths in operator subtrees**. In *Advances in Information Retrieval - 41st European Conference on IR Research, ECIR 2019, Cologne, Germany, April 14-18, 2019, Proceedings, Part I*, volume 11437 of *Lecture Notes in Computer Science*, pages 116–129. Springer.
- Chuyue Zhou, Hongya Lyu, Xinle Lin, Hengyu Zhao, Junshao Guo, Xueren Zhang, Shuaikang Xue, Qiang Ma, Jie Zhou, Yudong Wang, and Zhiyuan Liu. 2026. **UltraData-Math**. Hugging Face dataset, <https://huggingface.co/datasets/openbmb/UltraData-Math>.
- Fan Zhou, Zengzhi Wang, Nikhil Ranjan, Zhoujun Cheng, Liping Tang, Guowei He, Zhengzhong Liu, and Eric P. Xing. 2025. **MegaMath: Pushing the limits of open math corpora**. *Preprint*, arXiv:2504.02807.
- Justin Zobel. 1998. **How reliable are the results of large-scale information retrieval experiments?** In *SIGIR '98: Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, August 24-28 1998, Melbourne, Australia*, pages 307–314. ACM.

A Rubric Details

The judge returns one JSON object per pair. The schema constrains the fields jointly: a score of 2 requires `answer_validity = correct`; a score of 1 requires `correct` or `minor_error`; a score of 0 requires `major_error`, `uncertain` or `not_applicable`; an unevaluable query has no score, `validity not_applicable` and a code

from the last group below. Table 9 lists the reason codes allowed with each score. The prompt is prompts/judge_prompt.md (SHA-256 031ac72b649ae979. . .); guideline_version is always strict-math-retrieval-v1, so the hash, not the version string, identifies the prompt text.

Score	Allowed reason codes
2	full_solution, equivalent_solution, correct_short_answer
1	correct_partial_method, correct_partial_subquestion, minor_math_error
0	major_math_error, minor_math_error, wrong_problem, condition_mismatch, target_mismatch, topic_only, question_restatement, unresolved_attempt, contaminated_or_spliced, uncertain_math, missing_context, figure_dependent, degenerate_text
unevaluable	at least one of missing_context, figure_dependent, degenerate_text

Table 9: Reason codes allowed with each score.

B Construction Pipelines

Corpus step	Documents
Upstream pages (UltraData-Math L2)	13,707,851
Drop difficulty score 0–1	13,514,716
Exact dedup of raw page text	6,029,052
Content extraction (empty skipped)	5,929,019
Remove each page’s own question	5,929,019
Normalise, drop empty, exact dedup	4,340,031
Near-duplicate removal (MinHash + LLM review)	3,676,820
Remove degenerate and multi-problem documents	3,525,546

Table 10: Corpus construction. Question removal empties 816 documents, which the next step drops.

Declined pages. The query extractor declines 14.9% of documents: reference pages only (39.5% of declines), corrupted text (24.8%), a missing figure or context (19.6%), not mathematics (8.5%) and no problem stated (7.7%).

Near-duplicate review. MinHash uses 128 permutations over word 5-grams after stripping $\displaystyle$ and NFKC normalisation, with two LSH banding schemes (16 bands $\times$ 8 rows and

Query step	Queries
Non-empty extracted queries	5,044,326
Exact dedup of normalised text	3,385,028
Remove lexical overlap with evaluation queries	3,373,109
Remove semantic overlap with evaluation queries	3,357,494
Remove queries from multi-problem pages	2,911,860
Source page judged	2,901,088
Source page judged a full answer	2,032,033
Source page removed as degenerate or multi-problem	2,017,420
Source page removed as a near-duplicate, no full answer left	2,008,802
Remove queries that are not self-contained	1,853,431
Exhaustive lexical leakage check	1,853,389

Table 11: From extracted queries to training queries. When duplicates are merged, the source page kept is chosen at random rather than by shard order, which tracks crawl time and source site.

32 bands $\times$ 4 rows). The 762,610 proposed deletions split into 583,077 whose document has no content that the kept copy lacks, which are deleted, and 179,533 that go to GPT-5.6-Sol; the reviewer was calibrated on 96 hand-checked items and got all 96 right, and a failed call counts as “keep”. Of the 177,431 proposals it reviewed, it kept 54.8%: 49.9% had substantive content the other copy lacks and 4.9% were different documents. Pairs close to the 0.80 threshold can escape, because LSH candidate generation is probabilistic; normalising more aggressively before hashing merged 0.026% more groups on a 120,000-document sample, all of them true duplicates.

C Candidate-Generation Probes

Order	top 5	top 10	top 20
Dense only	1.61	2.56	3.99
Best rank across routes	1.78	2.70	4.03
Weighted RRF	1.92	2.97	4.56

Table 12: Full answers per query on a 1,000-query probe whose three-route top-100 union was judged in full. Weighted RRF uses weight 2 for the dense route and $k = 60$. Fusing by best rank lets documents that BM25 ranks high but that are not relevant push good dense results down; weighting the dense route keeps 97.8% of the fused top 5 inside the dense top 5 while still adding positives only the other routes find.

The reranker that selects the LLM-judged documents within dense ranks 6–20 was chosen on

a 196-query probe whose three-route pool was judged in full: Qwen3-Reranker-4B separates score 2 from score 0 with an AUC of 0.921, against 0.599 for the fused retrieval order.

D The 8B Relevance Model

Step	Within-query R^2	Agree	2 kept	0 $\leftrightarrow$ 2
5,000	.328 [.312, .344]			
10,000	.364 [.349, .380]	73.2	63.3	1.11
12,500	.371 [.356, .387]	73.6	64.6	1.10
15,000	.373 [.358, .389]	73.7	67.0	1.12

Table 13: The 8B model on the held-out set (31,830 pairs, 6,000 queries) during training; percentages use the cuts 0.5 and 1.5. The 10,000-step checkpoint labels the data; later steps add at most 0.5 points of agreement.

LLM label	8B $\rightarrow$ 0	8B $\rightarrow$ 1	8B $\rightarrow$ 2
0	75.4%	23.1%	1.4%
1	17.1%	76.5%	6.4%
2	2.4%	34.3%	63.3%

Table 14: How the 8B model’s cut labels spread within each LLM label on the held-out set, at 10,000 steps.

Scoring. Pairs are packed into rows of up to 65,536 tokens with position ids restarting at each pair, and scored with FlashAttention-2 (Dao, 2024) variable-length attention in bf16. Before producing any output, each worker scores 256 held-out pairs and compares them with the training-time predictions; it must reach a Pearson correlation of 0.999, and the measured value was 1.00000.

E Descriptive Statistics

	Min	Mean	Max
Document length	2	7,418	168,093
Training query length	3	200	8,894
Test query length	13	207	2,624
Candidates per training query	1	23.5	41
Relevant docs per test query	1	13.6	140

Table 15: Descriptive statistics computed from the released tables. Lengths are in characters (Unicode code points); candidates are judged pairs. 0.07% of documents are shorter than 100 characters, mostly short fragments.

The release is a Hugging Face dataset with four configurations. `corpus` holds `did`, `text`

and `n_chars`. `queries` holds `qid`, `text`, the `source_did` the query was extracted from and `from_page_question`. `judgments` holds one row per judged training pair with its rank in each retrieval route, its fusion score and stage, the label, the judge, the 8B raw score, the dense and reranker scores, and the LLM judge’s validity, reason codes, confidence and rationale. `qrels` lists the relevant documents of each test query. Document ids are “d_” plus the first 24 hex digits of the SHA-256 of the lowercased normalised text; training query ids hash the raw upstream page, and test query ids hash the normalised query text.

F Queries in Numbers

Of the 1,853,389 training queries, 797,856 (43.0%) are a page’s own question, minimally edited, and the rest were composed by the extractor. 64,544 training queries have no `source_did`: near-duplicate removal took their source page out, and its surviving copy is not a full answer or none was found, but another document is a full answer. On 2,901,088 judged source pages, the LLM judge gave 70.04% a 2 and 11.70% a 0.